

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Water = $65.0 - 55.0 = 10.0$ g; fresh soil = 40.0 g • $10.0 / 40.0 \times 100 = 25.0\%$	2
01.2	• Organic matter = $55.0 - 52.6 = 2.4$ g; dry soil = 30.0 g • $2.4 / 30.0 \times 100 = 8.0\%$	2
01.3	• So all water is removed and the mass loss on burning is due to organic matter only • Heating until mass no longer changes shows all water has evaporated	2
01.4	• Organic matter releases nutrients as it decomposes and provides food for soil biota • Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration • Surface organic matter / humus adhesion reduce rain splash and wind erosion	3
01.5	• Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
02.1	• $\Sigma n(n-1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ • $N(N-1) = 100 \times 99 = 9900$ • $D = 9900 / 2900 = 3.41$ (accept 3.4)	3
02.2	• Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
03.1	- Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium - Positive feedback: a change causes other changes that increase the original change	2
03.2	- Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced - Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less	4
03.3	- Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground - Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming - Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature - If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming	4
03.4	<i>Any 3 from:</i> - Reduced albedo as snow / ice melts - Ocean acidification reducing coral survival and carbon sequestration - Increased forest and peat fires releasing CO₂ - Faster decomposition of soil organic matter releasing CO₂ - Increased water vapour (a greenhouse gas) in warmer air - Increased cirrus (high) cloud which traps infrared	3
03.5	- A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable - So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
04.1	- Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged - Electrons flow through an external conductor / circuit to the relatively positive lower layer; the current powers appliances	2
04.2	- Smooth cells reflect about 30% of the light; grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away - Multi-junction cells have several layers of different materials, each absorbing different wavelengths - So a greater proportion of the available light is absorbed and converted to electricity	3
04.3	- Solar farm: polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m² matters less) - Satellite: multi-junction - highest efficiency where mass / area is critical and cost is less important - Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit	3

Question	Marking guidance	Marks
04.4	- Intermittent: no output at night and low in winter (daily and seasonal cycles) - predictable - Unpredictable: cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
05.1	C	1
05.2	- Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms - Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas	2
05.3	- HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere - So most do not persist long enough to reach the stratosphere where UV would release their chlorine	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
06.1	The dose that kills 50% of the (test) population	1
06.2	- Insecticide X: LD₅₀ for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well - Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) - Y has a much higher LD₅₀ for the beetle (150) so few predators killed / more specific	3
06.3	Any 2 from: - High water solubility so it is leached into rivers / groundwater / runoff - Toxic to aquatic organisms downstream - Mobile so it can affect areas away from the field - Washed off the crop quickly so more frequent application needed	2
06.4	- Long half-life = high persistence so it is not broken down quickly - So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble)	2
06.5	Any 2 from: - Absorbed and translocated within the plant so protects new growth - Not washed off by rain - Does not need to cover all leaf surfaces / less spray needed - Less spray drift so less harm to non-target species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	
07.1	- Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm⁻³ at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm⁻³ at 4 and 28 h) - The night-time minimum is lower on the second night (2.5 vs 3.5)	2

Question	Marking guidance	Marks
07.2	- Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) - During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) - At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen - Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen; warm summer water holds less oxygen and decay is rapid	4
07.3	- At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate - At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation	2
07.4	Any 2 from: - Cyanobacteria release toxins harmful to livestock, pets and people - Floating algae shade submerged / rooted macrophytes which die, disrupting food webs - Algae are an unreliable food source (die off rapidly) so food webs are poor - Blocked filters / unpleasant taste and smell in water supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	
08.1	B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
08.2	- Chemicals cause the oil molecules to join together / solidify - Producing more solid material that can be collected / lifted more easily than liquid oil	2
08.3	- Air contains about 210 cm^3 of oxygen per litre whereas water at 20°C holds a maximum of about 6 cm^3 per litre (less than 3% of the amount in air) - So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
09.1	$180 / 1.2 = 150$	1
09.2	Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both	1
09.3	- Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium ($1.2 \text{ vs } 0.05 \text{ mg kg}^{-1}$) - Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years - Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) - Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation)	4

Question	Marking guidance	Marks
09.4	<i>Any 2 from:</i> • Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble • Cover / cap spoil heaps to prevent leaching and dust • Recycle nickel-cadmium batteries / hazardous waste landfill • Phytoremediation of contaminated paddy soils	2
09.5	• Manufacture may release cadmium in effluents / dust • Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
10.1	• No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m • Reduction = 2.2 m	2
10.2	• After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream • During dry periods the dam is opened to release water and maintain river levels	2
10.3	• Deforestation / urbanisation reduce infiltration and interception • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects • Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) • Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use; life-cycle / embodied energy • Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25